

# Unified F\_U\_Bi for SMBH Binary Merger Dynamics: Inside-to-Outside and Outside-to-Inside

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## PAPER\_1104: Unified $F_{U,Bi}$ for SMBH Binary Merger Dynamics

### Abstract

We develop the complete buoyancy-mediated unified field theory for supermassive black hole (SMBH) binary mergers, distinguishing inside-to-outside (inspiral  $\rightarrow$  ringdown) and outside-to-inside (environment  $\rightarrow$  binary) dynamics. The two directional  $F_{U,Bi}$  components are derived from the 26-layer UQFF gravity framework with gravitational wave phonon coupling and chirp mass scaling.

### 1. Inside-to-Outside: Inspiral to Ringdown

The  $F_{U,Bi}$  component radiating outward during the merger:

$$F_{U,Bi}^{\text{in} \rightarrow \text{out}} = \sum_{i=1}^{26} [\text{Ug}1_i + \text{Ug}2_i + \text{Ug}3_i + \text{Ug}4_i] \times \left(1 + \frac{\Phi}{F_U} \cdot \Delta E_{\text{gw}}\right) \times \left(\frac{\mathcal{M}_c}{M_{\text{total}}}\right)^{5/3}$$

### 26-Layer Gravity Terms

Each layer  $i \in [1, 26]$  has quantum state factor  $Q_i = 1/(1 + \kappa i)$ :

| Term             | Expression                                                      | Physics              |
|------------------|-----------------------------------------------------------------|----------------------|
| Ug1 <sub>i</sub> | $Q_i \cdot \mu_s \nabla(M_s/r)$                                 | DPM-emergent gravity |
| Ug2 <sub>i</sub> | $(1 - [\text{SSq}])Q_i \cdot \mu_s \nabla(M_s/r) \times 0.1$    | Charge-reactivity    |
| Ug3 <sub>i</sub> | $[\text{SSq}] Q_i \cdot \mu_s \nabla(M_s/r) \times 0.05 \sin i$ | String rotation      |
| Ug4 <sub>i</sub> | $\kappa Q_i \times 10^{-15}$                                    | Vacuum concentration |

### Chirp Mass Factor

For a binary with masses  $M_1$  and  $M_2$ :

$$\mathcal{M}_c = M_{\text{total}} \cdot \eta^{3/5}, \quad \eta = \frac{M_1 M_2}{M_{\text{total}}^2}$$

$$\left(\frac{\mathcal{M}_c}{M_{\text{total}}}\right)^{5/3} = \eta$$

## GW Phonon Coupling

$$\Delta E_{\text{gw}} = \frac{GM_1 M_2}{2r_{\text{sep}}}$$

The phonon-GW coupling factor:

$$1 + \frac{\Phi(\omega, \Gamma)}{F_U} \cdot \Delta E_{\text{gw}}$$

## 2. Outside-to-Inside: Environmental Buoyancy

The complementary  $F_{U, Bi}$  component from the environment:

$$F_{U, Bi}^{\text{out} \rightarrow \text{in}} = F_U - F_{U, Bi}^{\text{in} \rightarrow \text{out}} - \sum_k F_{\text{tidal}, k}(r_k, M_k)$$

where  $F_{\text{tidal}, k}$  are tidal forces from surrounding structures (galaxy clusters, dark matter halos, accretion disk).

## 3. Net Buoyancy Imbalance

$$\Delta F = F_{U, Bi}^{\text{in} \rightarrow \text{out}} - F_{U, Bi}^{\text{out} \rightarrow \text{in}}$$

- $\Delta F > 0$ : merger-dominated (radiative outflow exceeds environmental inflow)
- $\Delta F < 0$ : environment-dominated (accretion exceeds radiation)
- $\Delta F = 0$ : buoyancy equilibrium

## 4. Merger Timescale

Using the Peters formula for gravitational wave inspiral:

$$t_{\text{merger}} = \frac{5}{256} \frac{c^5 a_0^4}{G^3 M_1 M_2 M_{\text{total}}}$$

## 5. Implementation

Calculator: `UnifiedFUBiSMBHMergerDynamicsCalculator` in `CondensedPhysics.py`

- Inputs:  $M_1$ ,  $M_2$ ,  $r_{\text{sep}}$ ,  $F_U$ , tidal forces, linewidth  $\Gamma$
- Outputs: both  $F_{U, Bi}$  components, net imbalance, chirp mass, merger timescale

## 6. Conclusion

The directional decomposition of  $F_{U,Bi}$  during SMBH mergers provides a complete description of buoyancy flow in the UQFF framework, with the inside-to-outside component governed by chirp mass scaling and GW phonon coupling, and the outside-to-inside component determined by the global unified field budget less tidal contributions.

## References

- Peters, P.C. (1964). Gravitational radiation and the motion of two point masses. *Phys. Rev.* 136, B1224.
- Murphy, D.T. (2025). UQFF: Star Cradle Mechanics framework. Star-Magic repository.